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三角関数の不定積分 ～ 分数型 ～

(例) 次の不定積分を求めよ。

Cは積分定数とする。

$$(1) \int \frac{1}{\cos x} dx = \int \frac{\cos x}{\cos^2 x} dx \\ = \int \frac{\cos x}{1 - \sin^2 x} dx$$

そこで、 $\sin x = t$ とおくと

$$\cos x dx = dt$$

であるから

$$\begin{aligned} (\text{与式}) &= \int \frac{1}{1 - \sin^2 x} \cdot \cos x dx \\ &= \int \frac{1}{1 - t^2} dt \\ &= \frac{1}{2} \int \left(\frac{1}{1-t} + \frac{1}{1+t} \right) dt \\ &= \frac{1}{2} (-\log|1-t| + \log|1+t|) + C \\ &= \frac{1}{2} \log \left| \frac{1+t}{1-t} \right| + C \\ &= \frac{1}{2} \log \left| \frac{1+\sin x}{1-\sin x} \right| + C \quad \leftarrow |1 \pm \sin x| > 0 \end{aligned}$$

$$\begin{aligned} (2) \int \frac{1}{1 + \cos x} dx &= \int \frac{1 - \cos x}{1 - \cos^2 x} dx \\ &= \int \frac{1 - \cos x}{\sin^2 x} dx \\ &= \int \left\{ \frac{1}{\sin^2 x} - (\sin x)' (\sin x)^{-2} \right\} dx \\ &= -\frac{1}{\tan x} + \frac{1}{\sin x} + C \end{aligned}$$

(別解)

$$\begin{aligned} \int \frac{1}{1 + \cos x} dx &= \int \frac{1}{2 \cos^2 \frac{x}{2}} dx \\ &= \tan \frac{x}{2} + C \end{aligned}$$

$$\begin{aligned} (3) \int \frac{1}{1 - \sin x} dx &= \int \frac{1 + \sin x}{1 - \sin^2 x} dx \\ &= \int \frac{1 + \sin x}{\cos^2 x} dx \\ &= \int \left\{ \frac{1}{\cos^2 x} + (\cos x)' (\cos x)^{-2} \right\} dx \\ &= \tan x + \frac{1}{\cos x} + C \end{aligned}$$

(別解)

$$\begin{aligned} \int \frac{1}{1 - \sin x} dx &= \int \frac{1}{1 - \cos(\frac{\pi}{2} - x)} dx \\ &= \int \frac{1}{2 \sin(\frac{\pi}{4} - \frac{x}{2})} dx \\ &= \frac{1}{\tan(\frac{\pi}{4} - \frac{x}{2})} + C \end{aligned}$$